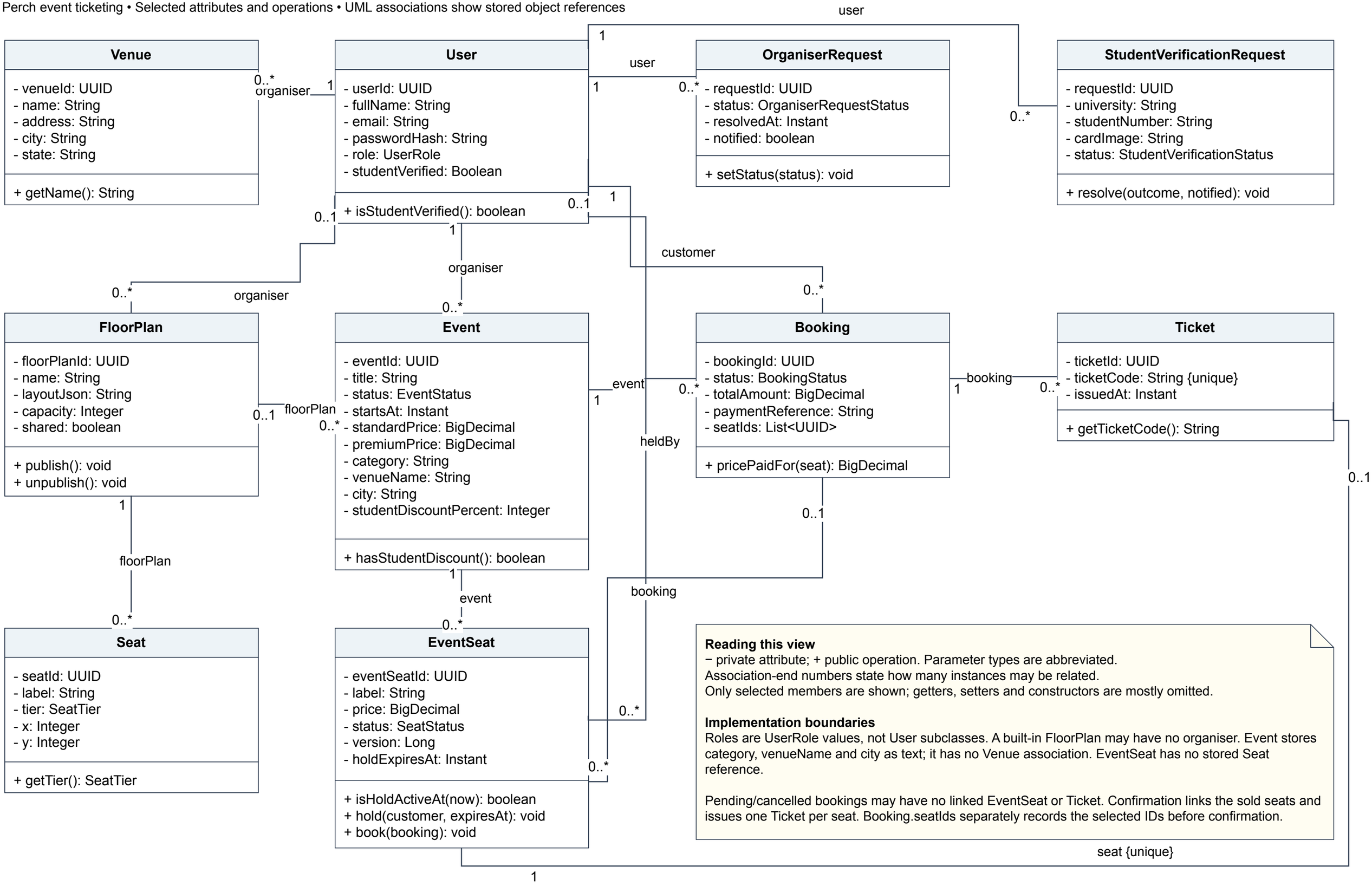
Class diagram | A Domain structure

Perch event ticketing • Selected attributes and operations • UML associations show stored object references



Class diagram | B Application structure

Perch event ticketing • Business operations and selected collaborators • Dashed arrows denote dependencies

UserService

- userRepository: UserRepository

+ register(…): UserDTO
+ login(…): UserDTO

OrganiserRequestService

+ submit(…): UserDTO
+ approve(…): OrganiserRequestDTO
+ reject(…): OrganiserRequestDTO

StudentVerificationService

+ submit(…): StudentVerificationDTO
+ approve(…): StudentVerificationDTO
+ reject(…): StudentVerificationDTO

AdminService

- eventSeatRepository: EventSeatRepository

+ getStats(): AdminStatsDTO

VenueService

+ getVenuesForOrganiser(…): List<VenueDTO>
+ getVenueForOrganiser(…): VenueDTO

EventService

+ createEvent(…): EventDTO
+ publishEvent(…): EventDTO
+ updateEventPricing(…): EventDTO
+ createFloorPlan(…): FloorPlanDTO
+ cloneSharedFloorPlan(…): FloorPlanDTO

SeatHoldService

+ holdSeat(…): SeatHoldDTO
+ releaseHold(…): void
+ releaseExpiredHolds(): int

ExpiredSeatHoldReleaseJob

+ releaseExpiredHolds(): void

BookingService

- paymentProvider: PaymentProvider

+ startCheckout(…): BookingDTO
+ confirmPaidBooking(…): BookingDTO
+ markPaymentFailed(…): void

TicketService

+ issue(…): void
+ getTickets(…): List<BookingDTO.Ticket>
+ qrPng(…): byte[]

BookingEmailService

- mailSender: JavaMailSender [0..1]

+ sendBookingConfirmation(…): void

<<interface>>
PaymentProvider

+ createPaymentIntent(…): PaymentIntentResult
+ autoConfirms(): boolean

MockPaymentProvider

+ autoConfirms(): boolean

StripePaymentProvider

- client: StripeClient

+ createPaymentIntent(…): PaymentIntentResult

Scope
All services and the scheduled job are shown. Members are selective; … abbreviates parameters.

Controllers delegate to services; repositories persist the domain objects in panel A. DTOs cross HTTP boundaries.

Hollow triangles denote interface realisation. Mock is the default payment mode. Stripe confirms by webhook.

JavaMailSender is optional. Email is skipped without mail configuration.